EXPLAINING FEATURE SPARSITY THROUGH THE MARGIN–REGULARIZATION TRADE-OFF

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ABSTRACT

Explaining the number of features in a network has been a long-desired result in mechanistic interpretability. Morwani et al. (2024) laid the baseline with their principle of margin-maximization, showing provably that in networks where the regularization ($L_{2,\nu}$) matches its homogeneity degree (ν), all features would be used for modular addition or group composition tasks. Empirically, however, networks trained with ordinary weight decay consistently show a sparse selection of features. This paper investigates why. Our inquiry focuses specifically on modular addition and group composition tasks in polynomial and multilinear models. Our primary theoretical finding centers on the cost of new features—a trade-off between the logit margin of the solution and regularization. While more features improve margin, they cost the model in regularization. Hence we show that an optimal number of features can be calculated for networks based on the task, their homogeneity degree and the regularization. Further, our theory explains the transition in the number of features from dense to sparse—as seen empirically in McCracken et al. (2025)—by showing that this transition arises on changing either the homogeneity degree or regularization of the network. Specifically, we empirically show that the normalized margin attained by these networks peaks at the predicted optimum for the network, matching the result of Morwani et al. (2024) for their specific case. The optimums are also consistent with the empirical results seen in Chughtai et al. (2023).

1 INTRODUCTION

Morwani et al. (2024) uses the pioneering principle of margin-maximization to enable the bounding of the margin of the network. However, their arguments remain limited to the specific regularization ($L_{2,\nu}$) and architecture (quadratic models) chosen and remain hard to extend to further architectures and to more conventional regularizations. Their theory shows that under their configuration, the margin maximal solution would use all frequencies (dense) in the solution.

However, as observed by Nanda et al. (2023) and Chughtai et al. (2023), under different architectures and regularizations, a sparse subset of frequencies (or irreducible representations) are utilized by the model. McCracken et al. (2025) similarly sees that under different architectures, the number of features transitions from dense to sparse.

This paper serves to generalize the *margin-maximization* principle introduced by Morwani et al. (2024) to arbitrary $L_q; 2 \leq q \leq \nu$ norms and across multilinear and polynomial architectures (homogeneity degree $\nu = n + 1$) in models that contain no intereference (no signal in the logits not dependent on the offset between a candidate and the correct answer).

This paper shows:

The explanation of the sparsity phenomenon and the optimal number of frequencies for a certain configuration

A global upper and lower bound on the normalized margin of models

A general scaling law for the optimal number of frequencies for a certain prime p

2 PRELIMINARIES

We consider one-hidden-layer networks with homogeneous neurons and no biases, $f(\theta, x) = \sum_{i=1}^{m} \phi(\omega_i, x)$ with $\theta = \{\omega_1, \ldots, \omega_m\}; \omega_i \in \mathbb{R}^p$, where $\phi(\lambda\omega, x) = \lambda^\nu \phi(\omega, x)$ for all $\lambda > 0$; ν is the *homogeneity degree*.

The $L_{2,b}$ norm is $\|\theta\|_{2,b} = \left(\sum_i \|\omega_i\|_2^b\right)^{1/b}$.

Modular addition For a prime $p > 2$ and $n \geq 1$ operands, the dataset is $D_p := \{((a_1, \ldots, a_n), \sum_1^k a_k \mod p) : a_k \in \mathbb{Z}_p\}$; for an input x, $y(x)$ is its label, and $f(\theta, x)[c]$ is the logit of candidate $c \in \mathbb{Z}_p$. Inputs are one-hot, so the embedding vector $u \in \mathbb{R}^p$ acts on operand a as $u(a)$. The *product neuron* is $\phi(\{u_1, \ldots, u_n, w\}, x) = w \prod_1^n u_n(a_n)$ and the *power neuron* is $w\left(\sum_1^n u_n(a_n)\right)^n$ with $\nu = n + 1$.

Fourier transform. For $v : \mathbb{Z}_p \to \mathbb{C}$, $\hat{v}(j) := \sum_k v(k)\rho^{-jk}$ with $\rho := e^{2\pi i/p}$. For real v, $\hat{v}(-j) = \overline{\hat{v}(j)}$, so the distinct non-zero frequencies are $F := \{1, \ldots, \frac{p-1}{2}\}$. The multidimensional transform of the logit table is $\hat{f}_\theta(k_1, \ldots, k_n; l) := \sum_{x,c} f(\theta, x)[c]\, \rho^{-(k_1 a_1 + \cdots + k_n a_n + lc)}$; for a product neuron it factorizes as $\hat{w}(l) \prod_1^n \hat{u}_n(k_n)$.

3 THEORETICAL APPROACH

Maximum margin As in Morwani et al. (2024), the margin of a datapoint x where the correct answer is y is $g(\theta, x, y) = f(\theta, x)[y] - \max_{y' \neq y} f(\theta, x)[y']$. The margin of the dataset is $h(\theta) = \min_{(x,y)\in D_p} g(\theta, x, y)$, and the maximum margin under $L_{2,b}$ norm is $\gamma_b^* := \max_{\|\theta\|_{2,b} \leq 1} h(\theta)$. By homogeneity the normalized margin of any θ is $h(\theta)/\|\theta\|_{2,b}^\nu$. By Wei et al. (2019, Theorem 4.1), the global minimizers of cross-entropy plus $\lambda\|\theta\|_{2,b}^b$ attain normalized margin tending to γ_b^* as $\lambda \to 0$. Hence, we study maximum margin solutions as ideal, theoretically achievable solutions for the network, without considering empirical achievability.

3.1 AVERAGED MARGINS AND FOURIER AMPLITUDES

The minimum margin is at most any averaged class-weighted margin (Morwani et al., 2024, Section 3). We average over D_p and put all weight on the candidate with offset δ from the answer. Let

$$\eta_\theta(\delta) := \mathbb{E}_x\big[f(\theta, x)[y(x) - \delta]\big], \qquad \delta \in \mathbb{Z}_p,$$

Lemma 1 (Average Margin)**.** *For every θ and $\delta \neq 0$, $h(\theta) \leq \eta_\theta(0) - \eta_\theta(\delta)$.*

Proof. For every x, $y(x) - \delta$ is an incorrect candidate, so $f(\theta, x)[y(x)] - f(\theta, x)[y(x) - \delta] \geq h(\theta)$. Average over x. □

As a result of the averaging, η_θ keeps only the part of the logit table that depends on the offset between label and candidate. The rest does not depend on the accuracy of the candidate itself and hence, it is termed *interference*. Lemma 1 says interference cannot increase the margin as it uses the average over every input.

η_θ is a real function on $\mathbb{Z}_p$, hence a sum of cosines (Appendix A.1):

$$\eta_\theta(\delta) = \bar{\eta}_\theta + \sum_{j \in F} \mathrm{Re}\big(\beta_j \rho^{j\delta}\big), \qquad \beta_j := 2p^{-(n+1)} \hat{f}_\theta(j, \ldots, j; -j). \tag{1}$$

We call $A_j := |\beta_j|$ the *amplitude* of frequency j; a frequency with $A_j > 0$ is a *feature*, as in Nanda et al. (2023); Morwani et al. (2024). Since f is a sum over neurons, $\beta_j = \sum_i \beta_{ij}$ with β_{ij} computed from neuron i alone; for a product neuron $\beta_{ij} = 2p^{-\nu}\overline{\hat{w}_i(j)} \prod_{t=1}^n \hat{u}_{it}(j)$. Frequency j separates the label from the candidate δ away by the *coverage* $\kappa_j(\delta) := 1 - \cos(2\pi j\delta/p)$, which satisfies $\sum_{\delta \neq 0} \kappa_j(\delta) = p$ and is small where $j\delta$ is near a multiple of p. Hence, several frequencies are needed to cover every offset to maximize margin.

Lemma 2. *For every θ and $\delta \neq 0$, $h(\theta) \leq \sum_{j\in F} \mathrm{Re}(\beta_j)\,\kappa_j(\delta) \leq \sum_{j\in F} A_j\,\kappa_j(\delta)$.*

Proof. Average Lemma 1 at δ and $-\delta$ and use equation 1. □

The margin is thus bounded from above by the amplitudes of the frequencies alone, and hence, is largest when each β_j is real and positive and each frequency contributes a cosine peaked at the label.

3.2 THE BUDGET-SPLITTING PROBLEM

For a model with homogeneity degree ν and regularization $L_{2,b}$ for modular addition over prime p, $s := \nu/b \in [1, \nu/2]$.

Every feature in a model is a product of ν weights and cost the model in regularization through $L_{2,b}$ norm.

For a product neuron, the amplitude A is proportional to the product of the weights. Hence, the minimum penalty (sum of the weights) to achieve amplitude A is when they are exactly equal by the AM-GM inequality.

The least penalty to achieve amplitude A is $A^{b/v} = A^{1/s}$.

In the model, we split an amplitude a_j over all the frequencies in the model over a unit cost budget $sum_j a_j^{1/s} \leq 1$.

For a maximizer y^*, $K^*(s,p) = |\mathrm{spt}(y^*)|$ for the number of frequencies.

$$\Gamma_s(p) := \max\Big\{ \min_{\delta\neq 0} \sum_{j\in F} y_j\,\kappa_j(\delta) \;:\; y \in \mathbb{R}^F_{\geq 0},\ \sum_{j\in F} y_j^{1/s} \leq 1 \Big\}. \tag{2}$$

Proposition 3 (coverage bound)**.** *With $\|A\|_{1/s} := \big(\sum_j A_j^{1/s}\big)^s$, every θ satisfies $h(\theta) \leq \Gamma_s(p)\,\|A\|_{1/s}$.*

Proof. If $A = 0$, Lemma 2 gives $h(\theta) \leq 0$. Otherwise $y := A/\|A\|_{1/s}$ is feasible for equation 2; apply Lemma 2 and multiply by $\|A\|_{1/s}$. □

In words: the margin is at most the amount spent on features times the best return per unit spent. It remains to price amplitudes in norm, which, as in Morwani et al. (2024), is a single-neuron optimization. Let $\beta_j(\omega)$ be the coefficient of the single neuron $\phi(\omega,\cdot)$ and

$$C_b := \max_{\|\omega\|_2\leq 1} \sum_{j\in F} |\beta_j(\omega)|^{b/\nu},$$

so that, by homogeneity, $\sum_j |\beta_j(\omega)|^{b/\nu} \leq C_b\|\omega\|_2^b$ for every neuron. For the product neuron $C_b^{\nu/b} = B_p := 2(2p\nu)^{-\nu/2}$ for all $2 \leq b \leq \nu$, and for the power neuron $C_b^{\nu/b} = n!\,B_p$ (Appendix C).

Theorem 4 (universal bound)**.** *For any one-hidden-layer network with homogeneity constant ν, any width, and any $2 \leq b \leq \nu$, $\gamma_b^* \leq C_b^{\nu/b}\,\Gamma_{\nu/b}(p)$.*

Proof. Let $r = b/\nu \leq 1$. As $t \mapsto t^r$ is subadditive and $A_j \leq \sum_i |\beta_{ij}|$, $\sum_j A_j^r \leq \sum_{i,j} |\beta_{ij}|^r \leq C_b\|\theta\|_{2,b}^b$. Raise to the power $1/r$ and apply Proposition 3. □

No hypothesis on the neurons is used: they may mix frequencies, carry interference, or be inactive. Subadditivity says that splitting a feature over several neurons never makes it cheaper.

Corollary 5 ($b = \nu$: every frequency is used)**.** *$\Gamma_1(p) = p/(p-1)$, with unique maximizer $y_j = 2/(p-1)$ for all j, so at $b = \nu$ any network attaining Theorem 4 has equal amplitude on every frequency. For the quadratic network of Morwani et al. (2024) the bound is $2B_p\frac{p}{p-1} = \sqrt{2/27}\,/\,(p^{1/2}(p-1))$, exactly their maximum $L_{2,3}$-margin. (Proof: Appendix A.2.)*

For $b < \nu$ the bound holds but is not attained: a neuron that carries a frequency also carries interference at that frequency. The next section prices its removal.

4 MODULAR ADDITION

Interference at the feature's own frequency. A product neuron builds $\cos(2\pi j(a_1 + \cdots + a_n - c)/p)$ by multiplying one cosine per operand, and the product-to-sum formulas also produce *sign-flipped* copies: for $n = 2$, $\cos\frac{2\pi ja}{p}\cos\frac{2\pi jb}{p} = \frac{1}{2}\cos\frac{2\pi j(a+b)}{p} + \frac{1}{2}\cos\frac{2\pi j(a-b)}{p}$, and the second term does not depend on the label. In general, for each non-empty $T \subseteq [n]$ the neuron carries the cosine of $2\pi j(\sum_{t\notin T} a_t - \sum_{t\in T} a_t - c)/p$, interference at the feature's own frequency. We measure it by $\beta_{j,T} := 2p^{-(n+1)}\hat{f}_\theta(k_1, \ldots, k_n; -j)$ with $k_t = -j$ for $t \in T$ and $k_t = j$ otherwise, so $\beta_{j,\emptyset} = \beta_j$ and $\beta_{j,T} = \sum_i \beta_{ij,T}$.

Definition 6 (interference-free network)**.** A network is *interference-free* if $\beta_{j,T} = 0$ for every $j \in F$ and every non-empty $T \subseteq [n]$.

The condition is on the network's logit table, not its neurons: neurons may carry several frequencies and interference, provided it cancels in the sum. Interference at other modes is unrestricted; Lemma 1 discounts it and C_b prices it. For one neuron the copies are rigidly tied to the feature.

Lemma 7 (sign rule)**.** *For product and power neurons, for each neuron i and frequency j there are unit complex numbers $\xi_{i1j}, \ldots, \xi_{inj}$ with $\beta_{ij,T} = \beta_{ij}\prod_{t\in T}\xi_{itj}$ for every $T \subseteq [n]$. (Proof: Appendix C.)*

So a single neuron is never interference-free at a frequency it uses, and several neurons must cancel each other's copies. Relabel the neurons with $\beta_{ij} \neq 0$ as $l = 1, \ldots, N$, with unit factors ξ_{lt} and total $\beta = \beta_j$. The network is interference-free at j exactly when

$$\sum_{l=1}^{N} \beta_l \prod_{t\in T} \xi_{lt} \;=\; \beta\,[T = \emptyset] \qquad \text{for every } T \subseteq [n], \tag{3}$$

a finite system that involves neither p nor the task.

Lemma 8 (cost of a feature)**.** *Let $\nu = n + 1$, $|\xi_{lt}| = 1$, and let $\beta_1, \ldots, \beta_N$ satisfy equation 3. Then for every $r \in (0, 1]$, $\sum_l |\beta_l|^r \geq \nu^{1-r}|\beta|^r$. If $\beta = 0$, equality holds exactly when all $\beta_l = 0$. If $\beta \neq 0$ and $r < 1$, equality holds if and only if exactly ν of the β_l are non-zero, all of modulus $|\beta|/\nu$, with pairwise orthogonal chain vectors $m_l := (1, \xi_{l1}, \xi_{l1}\xi_{l2}, \ldots, \xi_{l1}\cdots\xi_{ln}) \in \mathbb{C}^\nu$.*

A non-zero clean feature requires at least ν contributing neurons; for $r < 1$, its least cost is attained by exactly ν equal-magnitude contributions, and more neurons never make it cheaper; the cancellation costs a factor ν^{1-r} over a single (interfering) neuron. At $r = 1$ the inequality is the triangle inequality, which is why the neuron count is pinned only for $b < \nu$.

Proof idea (Appendix B.1). For $\beta \neq 0$, the matrix $H = \sum_l \beta_l m_l m_l^*$ has below-diagonal entries equal to twisted sums of equation 3 over intervals of operands, hence zero, and diagonal β; so $\det(I + tH/\beta) = (1 + t)^\nu$. Expanding the determinant over subsets of neurons and bounding Gram determinants by one (Hadamard) gives $\prod_l(1 + x_l t) \geq (1 + t)^\nu$ with $x_l = \nu|\beta_l|/|\beta|$, and integrating the logarithm against $t^{-1-r}dt$ gives $\sum_l x_l^r \geq \nu$. □

Theorem 9. *Consider product networks with n operands ($\nu = n + 1$), let $2 \leq b \leq \nu$, $s = \nu/b$, and let y^* maximize $\Gamma_s(p)$. For width $m \geq \nu|\mathrm{spt}(y^*)|$, the maximum $L_{2,b}$-margin over interference-free networks on D_p is*

$$\gamma^*_{b,\mathrm{IF}} = \nu^{-(\nu-b)/b}\, B_p\, \Gamma_{\nu/b}(p).$$

For $b < \nu$, any interference-free network achieving it satisfies:

1. *its amplitudes are proportional to a maximizer $y^\star$ of equation 2, so it uses exactly $|\mathrm{spt}(y^\star)|$ frequencies;*
2. *every frequency it uses is carried by exactly ν neurons, with $|\beta_{ij}| = A_j/\nu$ and pairwise orthogonal chain vectors; no neuron carries an unused frequency.*

Proof outline (Appendix B.2). By Lemmas 7 and 8, $\sum_i |\beta_{ij}|^r \geq \nu^{1-r}A_j^r$ at each j ($r = b/\nu$); summing and applying the single-neuron budget gives $\nu^{1-r}\sum_j A_j^r \leq C_b\|\theta\|_{2,b}^b$, and Proposition 3

gives the bound. Attainment is Proposition 10. At equality, Proposition 3 is tight (condition 1) and Lemma 8 is tight at each frequency (condition 2). □

The class value is the universal bound divided by $\nu^{(\nu-b)/b}$ ($\sqrt{3}$, 4, $5^{3/2}$ at $\nu = 3, 4, 5$ under weight decay; 1 at $b = \nu$): the price of splitting each feature over ν neurons. The budget-splitting problem, and so the number of features, is the same in both results.

Proposition 10 (the ν-block)**.** *For $j \in F$, $\lambda > 0$ and $\alpha_l = \pi l/\nu$ ($l = 0, \dots, n$), the ν product neurons $u_t^{(l)}(a) = \lambda \cos(2\pi j a/p + \alpha_l)$, $w^{(l)}(c) = \lambda \cos(2\pi j c/p + n\alpha_l)$ have total logit table $\nu\lambda^\nu 2^{-n} \cos(2\pi j(a_1 + \cdots + a_n - c)/p)$ and $\|\omega^{(l)}\|_2^2 = \nu\lambda^2 p/2$. One block per $j \in \mathrm{spt}(y^*)$, with $\lambda_j^\nu = 2^n y_j^*/\nu$, attains Theorem 9. (Proof: Appendix B.3.)*

The phases are a rotated copy of the νth roots of unity, the equality case of Lemma 8: summed over the block, they cancel every sign-flipped copy. The weights $\lambda_j^\nu \propto y_j^*$ matter, since optimal shares are not equal.

The shape of the optimum. Theorem 9 reduces the optimal interference-free network to a maximizer of equation 2, and the maximizer's structure can be read off exactly. Rescaling a feasible y so that its worst coverage is one puts it in the polyhedron

$$\mathcal{P}_p := \Big\{ v \in \mathbb{R}^F_{\geq 0} : \ \textstyle\sum_{j \in F} v_j \, \kappa_j(\delta) \geq 1 \ \text{ for all } \delta \neq 0 \Big\},$$

and $\Gamma_s(p)^{-1/s} = \min_{v \in \mathcal{P}_p} \sum_j v_j^{1/s}$. The polyhedron depends on p only, not on ν, b or n: the exponent s decides which of its points is cheapest, and nothing else.

Proposition 11 (finitely many shapes)**.** *Let $s > 1$. Every maximizer of equation 2 is $y = \Gamma_s(p)\, v$ for a vertex v of $\mathcal{P}_p$ that minimizes $\sum_j v_j^{1/s}$. There is a finite set $E \subset (1, \infty)$ such that the set of minimizing vertices is constant on each component of $(1, \infty) \setminus E$. Vertices whose non-zero entries agree as multisets (such as images under multiplication by a non-zero residue, identifying j with $-j$) tie for every s. Along a fixed minimizing vertex branch $y = \Gamma_s(p)v$, the support and non-zero amplitude ratios are constant, and the budget shares $y_j^{1/s} = v_j^{1/s} / \sum_k v_k^{1/s}$ vary continuously. Shares may jump when the branch changes; an arbitrary choice among tied vertices need not be continuous or have fixed support. (Proof: Appendix B.4.)*

Thus the available optimal shapes change only at finitely many exponents; following one branch keeps its frequencies and amplitude ratios fixed until a switch. At $p = 13$, for $s \in (1, 5/2]$, the maximizer uses 6, 4, 3 and then 2 frequencies, switching at $s = 1.285$, 1.407 and 1.678. At $p = 23$ there are eight switches on $(1, 5/2]$; at one of them ($s = 1.955$) the three frequencies stay the same and only their ratios change.

Proposition 12 (bottleneck offsets)**.** *Let $s > 1$ and let y be a maximizer with $K = |\mathrm{spt}(y)|$. There are offsets $\delta_1, \dots, \delta_K \in \{1, \dots, \frac{p-1}{2}\}$ with $\sum_j y_j \kappa_j(\delta_i) = \Gamma_s(p)$ for every i, whose coverage vectors restricted to $\mathrm{spt}(y)$ are linearly independent. On its support, y is the unique solution of these K equations. In the network of Proposition 10, the gap between the label and the candidate δ below it is $\sum_j y_j \kappa_j(\delta)$ on every input, so the $2K$ candidates at offsets $\pm\delta_i$ from the label are, on every input, all hardest wrong answers at once. (Proof: Appendix B.4.)*

An optimal network is therefore held in place by at least as many equally hard wrong answers as it has features, and its amplitude ratios are exactly those that make these wrong answers equally hard. In every exact computation at $p \leq 23$ and $s = 3/2$ the number of such offsets is exactly K.

Proposition 13 (removing a feature)**.** *Let $s > 1$. Let y be feasible for equation 2 with $\sum_j y_j^{1/s} = 1$ and worst coverage $m \geq (1 - \varepsilon)\, \Gamma_s(p)$, where $\varepsilon \in [0, 1)$. For $j \in \mathrm{spt}(y)$, let $w_j := y_j^{1/s}$ be the share of the budget spent on j, and let m_{-j} be the worst coverage after setting $y_j = 0$. Then*

$$m - 2y_j \ \leq \ m_{-j} \ \leq \ \frac{(1 - w_j)^s}{1 - \varepsilon}\, m.$$

In particular, removing feature j from a maximizer costs at least the fraction $1 - (1 - w_j)^s$ of the margin, which lies between w_j and $s\, w_j$. (Proof: Appendix B.4.)

Table 1: $K^*(\nu/b, p)$ under weight decay ($b = 2$). Support enumeration is exhaustive for $K \leq 8, 7, 5$ at $p = 59, 71, 113$ and heuristic above.

p	$\nu = 3$ ($s = 3/2$)	$\nu = 4$ ($s = 2$)	$\nu = 5$ ($s = 5/2$)
59	5	4	3
71	5	4	3
113	7	4	4

In the network of Proposition 10, deleting the ν neurons of frequency j removes exactly $y_j \cos(2\pi j(a_1 + \cdots + a_n - c)/p)$ from the table, so these bounds apply to that network directly. Every feature is load-bearing in proportion to its share of the regularization budget. At exact maximizers with $p \leq 23$, the actual loss is 1.15–1.72, 1.04–1.41 and 1.00–1.18 times the lower bound at $s = 3/2$, 2 and $5/2$.

How many features. Under weight decay the count falls as the degree rises and grows only slowly with p (Table 1). The dense network of Corollary 5 reaches 0.52–0.63 of the interference-free optimum at $\nu = 3$, 0.15–0.25 at $\nu = 4$ and 0.04–0.08 at $\nu = 5$: the preference for few features sharpens with each step in degree. Whether a network carrying interference can beat the class value is open (Appendix D); trained networks do carry interference, which Section 6 measures.

How the count grows with p. For fixed $2 \leq b < \nu$, every programme maximizer uses $\Theta(\log p)$ frequencies. The same bounds also determine the order of the unrestricted maximum margin. For unrestricted networks, we obtain an $\Omega(\log p)$ requirement on the number of features in a constant-factor optimal network, but no matching support upper bound.

Theorem 14 (logarithmic scaling)**.** *Fix $s > 1$ and let $a_s := 3 + s/2$. For every odd prime p,*

$$\tfrac{1}{2}\lceil 8 \ln p \rceil^{1-s} \;\leq\; \Gamma_s(p) \;\leq\; \tfrac{3}{2}(2a_s)^s (\ln p)^{1-s}.$$

Every maximizer of equation 2 uses between $c_s \ln p$ and $9 \cdot 4^{1/(s-1)} \ln p$ frequencies, where $c_s = \big(8a_s^2(3 \cdot 9^{s-1})^{1/s}\big)^{-1}$, and spends a share of at least $4^{-1/(s-1)}/(9 \ln p)$ of its budget on each of them. Consequently, for product networks with $2 \leq b < \nu$, write $\gamma^*_{b,m}(p)$ for the maximum $L_{2,b}$-margin at width m, with interference allowed. Uniformly over $m \geq \nu\lceil 8 \ln p \rceil$,

$$\gamma^*_{b,m}(p) = \Theta_{\nu,b}\big(p^{-\nu/2}(\ln p)^{1-\nu/b}\big).$$

(Proof: Appendix B.5.)

The last statement needs no answer to the open question of Appendix D. The universal upper bound and a block construction using at most $\nu\lceil 8 \ln p \rceil$ neurons give the same dependence on p, with constants depending only on ν, b. If the width also realizes a programme maximizer, the sharper comparison is $\nu^{1-s} B_p \Gamma_s(p) \leq \gamma^*_{b,m}(p) \leq B_p \Gamma_s(p)$. Even with interference, a near-optimal network must use sufficiently many features:

Corollary 15 (few features cannot be optimal)**.** *Let $2 \leq b < \nu$, $s = \nu/b > 1$, $a_s = 3 + s/2$, and $K \geq 1$. Let $\theta \neq 0$ be any network (product or power, any width, interference allowed) whose amplitudes A_j are non-zero on at most K frequencies, and let $x := \ln p/K$. Then*

$$\frac{h(\theta)}{\|\theta\|^{\nu}_{2,b}} \;\leq\; C_b^{\nu/b} \cdot \tfrac{3}{2}\Big(\frac{x}{e^{x/(2a_s)} - 1}\Big)^s (\ln p)^{1-s}.$$

For fixed ν, b, product networks with $K = o(\log p)$ therefore attain a vanishing fraction of $\gamma^*_{b,m}(p)$ at any comparison width $m \geq \nu\lceil 8 \ln p \rceil$. (Proof: Appendix B.5.)

The constants are loose, about two orders of magnitude apart at $s = 3/2$. The theorem fixes the growth rate, not the count at a given p; Table 1 gives the counts.

The theorem also shows how narrow the dense regime is. By Corollary 5 every frequency is used at $b = \nu$. By the share floor of Theorem 14, however, for $(p-1)/2 > \lceil 8 \ln p \rceil$, a programme maximizer can use all $\frac{p-1}{2}$ frequencies only if $s - 1 \leq \ln 4 / \ln\big(\frac{p-1}{2\lceil 8 \ln p \rceil}\big) = O(1/\log p)$. So the range of b below ν on which the programme optimum stays dense shrinks as p grows. Computed exactly, the optimum is dense up to $s = 1.548$, 1.345, 1.285, 1.217, 1.194 and 1.160 at $p = 7$, 11, 13, 17, 19 and 23.

5 FINITE GROUPS

We conclude by studying group composition on finite groups G: the input space is $\mathcal{X} := G \times G$, the output space is $\mathcal{Y} := G$, and given inputs $a, b \in G$ the network is trained to predict $c = ab$. The same argument as in Sections 3 and 4 applies, with frequencies replaced by irreducible representations. It gives an exact margin ceiling in which each irreducible representation pays a price set by how cheaply the matrix-multiplication tensor can be written as a sum of products; at even degree this is the schoolbook algorithm, with d^ν neurons per d-dimensional representation. Among networks without same-representation interference this ceiling is attained and fixes the number of features: one representation on S_5 and the Galois-conjugate pair on A_5, as observed. Unlike on cyclic groups of prime order, networks with structured interference can exceed the ceiling on some groups (D_3, S_4, $\mathbb{Z}_{12}$), so there the count is exact only within the interference-free class.

6 EXPERIMENTS

6.1 WHERE INTERFERENCE ENTERS TRAINED NETWORKS

6.2 THE EXPONENT $s = \nu/b$

6.3 MOVING THE DEGREE AND THE NORM SEPARATELY

7 DISCUSSION

REPRODUCIBILITY STATEMENT

All theorems are stated with their hypotheses in Sections 3 and 4 and proved in full in the Appendix; the constants of every construction were checked numerically to machine precision. Experimental settings are listed in Section 6; code and configuration files are provided as supplementary material.

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Appendix organization. Appendix A proves the results of Section 3. Appendix B proves the results of Section 4: the cost of a feature (Lemma 8), Theorem 9, the ν-block (Proposition 10), the shape of the optimum (Propositions 11–13) and logarithmic scaling (Theorem 14, Corollary 15). Appendix C computes the single-neuron budget and checks the sign rule for the product and power networks. Appendix D states the conjecture that the interference-free value is the global maximum, and what is known about it.

Throughout, for $v : \mathbb{Z}_p \to \mathbb{R}$ we use $\hat{v}(j) = \sum_a v(a)\rho^{-ja}$, so that $v(a) = p^{-1}\sum_j \hat{v}(j)\rho^{ja}$, $\sum_{j\in\mathbb{Z}_p} |\hat{v}(j)|^2 = p\|v\|_2^2$ (Parseval), $\hat{v}(-j) = \overline{\hat{v}(j)}$, and hence $\sum_{j\in F} |\hat{v}(j)|^2 \le \frac{p}{2}\|v\|_2^2$.

A Proofs for Section 3

A.1 The spectrum of the averaged logit

Write the logit table through its multidimensional Fourier transform, $f(\theta, x)[c] = p^{-(n+1)} \sum_{k_1,\dots,k_n,l} \hat{f}_\theta(k_1, \dots, k_n; l)\, \rho^{k_1 a_1 + \cdots + k_n a_n + lc}$. Substituting $c = y(x) - \delta = a_1 + \cdots + a_n - \delta$ and averaging over x,

$$\eta_\theta(\delta) = p^{-(n+1)} \sum_{k,l} \hat{f}_\theta(k; l)\, \rho^{-l\delta} \prod_{t=1}^{n} \mathbb{E}_{a_t}\big[\rho^{(k_t+l)a_t}\big].$$

Each average is 1 if $k_t = -l$ and 0 otherwise, so only the modes $(j, \dots, j; -j)$ survive (with $j = -l$):

$$\eta_\theta(\delta) = p^{-(n+1)} \sum_{j\in\mathbb{Z}_p} G_j\, \rho^{j\delta}, \qquad G_j := \hat{f}_\theta(j, \dots, j; -j).$$

Since the table is real, $G_{-j} = \overline{G_j}$, and pairing j with $-j$ gives equation 1 with $\bar{\eta}_\theta = p^{-(n+1)} G_0$ and $\beta_j = 2p^{-(n+1)} G_j$. The averaged form used in Lemma 2 is

$$\eta_\theta(0) - \tfrac{1}{2}\big(\eta_\theta(\delta) + \eta_\theta(-\delta)\big) = \sum_{j\in F} \mathrm{Re}(\beta_j)\big(1 - \cos(2\pi j\delta/p)\big) = \sum_{j\in F} \mathrm{Re}(\beta_j)\, \kappa_j(\delta),$$

because $\mathrm{Re}(\beta\rho^{j\delta}) + \mathrm{Re}(\beta\rho^{-j\delta}) = 2\,\mathrm{Re}(\beta)\cos(2\pi j\delta/p)$. The same computation for one neuron defines β_{ij}. For a product neuron $\hat{f}$ factorizes, so $G_{ij} = \hat{u}_{i1}(j) \cdots \hat{u}_{in}(j)\, \hat{w}_i(-j)$, and $p^{-(n+1)} = p^{-\nu}$.

A.2 Proof of Corollary 5

For feasible y (at $s = 1$, $\sum_j y_j \le 1$),

$$\min_{\delta\neq 0} \sum_j y_j \kappa_j(\delta) \le \frac{1}{p-1} \sum_{\delta\neq 0} \sum_j y_j \kappa_j(\delta) = \frac{p}{p-1} \sum_j y_j \le \frac{p}{p-1},$$

using $\sum_{\delta\neq 0} \kappa_j(\delta) = p$. The flat choice $y_j = 2/(p-1)$ attains this, as we now show together with uniqueness. Equality forces $\sum_j y_j = 1$ and the coverage $\sum_j y_j \kappa_j(\delta)$ to be constant in $\delta \neq 0$, i.e. $\zeta(\delta) := \sum_{j\in F} y_j \cos(2\pi j\delta/p)$ equals a constant c_0 for $\delta \neq 0$, with $\zeta(0) = 1$. So $\zeta = c_0 + (1 - c_0)[\delta = 0]$. Its normalized Fourier coefficients $\frac{1}{p}\sum_\delta \zeta(\delta)\rho^{-k\delta}$ are $c_0 + (1 - c_0)/p$ at $k = 0$ and $(1 - c_0)/p$ at $k \neq 0$. From the cosine form they are 0 at $k = 0$ and $y_j/2$ at $k = \pm j$. Hence $c_0 = -1/(p-1)$ and $y_j = 2(1 - c_0)/p = 2/(p-1)$ for every j.

For the product network, attainment is Proposition 10 at $b = \nu$ with y^* flat. The equality conditions are those of Proposition 3 (flat amplitudes) and of the triangle inequality (at each j, all non-zero β_{ij} have the same phase). For the quadratic network of Morwani et al. (2024), $C_3 = 2B_p$ (Appendix C), and $2B_p \cdot \frac{p}{p-1} = 4(6p)^{-3/2}\frac{p}{p-1} = \frac{2}{3\sqrt{6}} \cdot \frac{1}{p^{1/2}(p-1)} = \sqrt{\frac{2}{27}} \cdot \frac{1}{p^{1/2}(p-1)}$. □

B Proofs for Section 4

B.1 Proof of Lemma 8

Lemma 16 (chain lemma)**.** *$\|m_l\|_2^2 = \nu$, and $H := \sum_l \beta_l m_l m_l^*$ is upper triangular with every diagonal entry equal to β.*

Proof. Index the chain coordinates by $0, \dots, n$. Every entry of m_l has modulus 1, so $\|m_l\|_2^2 = \nu$ and the diagonal of H is $\sum_l \beta_l = \beta$. For $t > t'$, the (t, t') entry of $m_l m_l^*$ is $\prod_{u=t'+1}^{t} \xi_{lu}$. Its sum against β_l vanishes by equation 3, applied to the non-empty interval $T = \{t'+1, \dots, t\}$. □

The entries above the diagonal need not vanish: H is triangular, not necessarily Hermitian. Triangularity is all the proof needs.

Proof of Lemma 8. If $\beta = 0$, the inequality is immediate, with equality exactly when every $\beta_l = 0$. Otherwise divide the coefficients by β; equation 3 is preserved with total 1. We prove the normalized claim and restore the factor $|\beta|^r$ at the end. For $r = 1$, the claim is simply $\sum_l |\beta_l| \geq |\sum_l \beta_l| = 1$. Henceforth let $0 < r < 1$.

Step 1: compare determinants. Put $u_l = m_l/\sqrt{\nu}$, $d_l = \nu\beta_l$ and $x_l = |d_l|$, so that the u_l are unit vectors and $H = \sum_l d_l u_l u_l^*$. Let $U = [u_1 \cdots u_N]$ and $\Delta = \operatorname{diag}(d_1, \dots, d_N)$. Sylvester's identity and expansion by principal minors give

$$\det(I_\nu + tH) = \det(I_N + t\Delta U^* U) = \sum_{L \subseteq [N]} t^{|L|} \Big(\prod_{l \in L} d_l \Big) G_L, \qquad G_L := \det \operatorname{Gram}(u_l : l \in L). \tag{4}$$

Gram positivity and Hadamard's inequality give $0 \leq G_L \leq 1$, with $G_L = 1$ exactly when the vectors in L are orthonormal. By Lemma 16, $\det(I + tH) = (1+t)^\nu$. Taking absolute values in equation 4 therefore yields, for $t \geq 0$,

$$(1+t)^\nu \leq \sum_{L \subseteq [N]} t^{|L|} \prod_{l \in L} x_l = \prod_{l=1}^{N} (1 + t x_l). \tag{5}$$

The same comparison coefficient by coefficient gives $e_k(x) \geq \binom{\nu}{k}$ for $0 \leq k \leq \nu$, where e_k is the kth elementary symmetric polynomial; in particular, at least ν coefficients are non-zero.

Step 2: integrate the logarithm. Define

$$J_r(x) := \int_0^\infty \log(1 + xt)\, t^{-1-r}\, dt, \qquad x \geq 0.$$

For $x > 0$ the integrand is $O(t^{-r})$ near zero and $O(t^{-1-r} \log t)$ near infinity, so the integral converges. Substitution gives $J_r(x) = x^r J_r(1)$, also valid at $x = 0$, with $0 < J_r(1) < \infty$. Integrating the logarithm of equation 5 against $t^{-1-r} dt$ and cancelling $J_r(1)$ gives

$$J_r(1) \sum_l x_l^r = \sum_l J_r(x_l) \geq \nu J_r(1), \qquad \text{hence} \qquad \sum_l x_l^r \geq \nu.$$

Restoring the original coefficients, $x_l = \nu|\beta_l|/|\beta|$, proves $\sum_l |\beta_l|^r \geq \nu^{1-r} |\beta|^r$.

Step 3: characterize equality. At equality, the continuous non-negative function $\sum_l \log(1 + x_l t) - \nu \log(1+t)$ has zero integral against a strictly positive density, so it vanishes for every $t > 0$. Thus $\prod_l (1 + x_l t) = (1+t)^\nu$ as polynomials. Their roots show that the non-zero x_l are exactly ν copies of 1. Let L_0 be their support. Comparing the coefficient of t^ν in equation 4 gives $1 = (\prod_{l \in L_0} d_l) G_{L_0}$. Since $|\prod_{l \in L_0} d_l| = 1$, we have $G_{L_0} = 1$: the corresponding u_l are orthonormal, and the chain vectors m_l are orthogonal. Conversely, exactly ν non-zero coefficients of modulus $|\beta|/\nu$ satisfying equation 3 give $\sum_l |\beta_l|^r = \nu^{1-r} |\beta|^r$. □

Remark 17 (orthogonality as roots of unity). Two chain vectors $m_l, m_{l'}$ are orthogonal if and only if $\sum_{t=0}^{n} \prod_{u \leq t} \xi_{l'u} \overline{\xi_{lu}} = 0$. If the ratios $\xi_{l'u} \overline{\xi_{lu}} = \zeta$ agree across operands, this reads $1 + \zeta + \cdots + \zeta^n = 0$: ζ is a non-trivial ν^{th} root of unity. This is the configuration of Proposition 10.

Remark 18 (other penalties). For $0 < r < 1$, $x \mapsto x^r$ is a positive integral of the functions $x \mapsto \log(1+xt)$. The same proof, together with $\sum_l x_l \geq \nu$, gives $\sum_l \psi(\nu|\beta_l|/|\beta|) \geq \nu\,\psi(1)$ for every $\psi(x) = cx + \int_0^\infty \log(1+xt)\,\mu(dt)$ with $c \geq 0$ and $\mu \geq 0$, whenever these integrals are finite and $\beta \neq 0$. The result is not special to power penalties.

B.2 Proof of Theorem 9

Let $r = b/\nu$, so $1/r = s$.

Upper bound. Let θ be interference-free. At each $j \in F$, the coefficients $(\beta_{ij})_i$ satisfy equation 3 with total β_j: for $T = \emptyset$ this is $\beta_j = \sum_i \beta_{ij}$, and for $T \neq \emptyset$ it is $0 = \beta_{j,T} = \sum_i \beta_{ij} \prod_{t\in T} \xi_{itj}$ by Lemma 7. Lemma 8 gives $\sum_i |\beta_{ij}|^r \geq \nu^{1-r} A_j^r$. Summing over j and using the single-neuron budget,

$$\nu^{1-r} \sum_j A_j^r \;\leq\; \sum_i \sum_j |\beta_{ij}|^r \;\leq\; C_b \sum_i \|\omega_i\|_2^b = C_b \|\theta\|_{2,b}^b. \tag{6}$$

Raising to the power $1/r$, $\|A\|_{1/s} \leq \nu^{(r-1)/r} C_b^{1/r} \|\theta\|_{2,b}^{b/r} = \nu^{-(\nu-b)/b} B_p \|\theta\|_{2,b}^\nu$, and Proposition 3 gives $h(\theta) \leq \nu^{-(\nu-b)/b} B_p \Gamma_s(p) \|\theta\|_{2,b}^\nu$.

Attainment. Proposition 10 gives an interference-free network with $\nu|\mathrm{spt}(y^*)|$ neurons attaining the value; any further neurons are set to zero.

Structure ($b < \nu$). Let θ be interference-free and attain the value. The value is positive, so $h(\theta) > 0$ and $A \neq 0$. The two inequalities $h(\theta) \leq \Gamma_s(p)\|A\|_{1/s}$ and $\|A\|_{1/s} \leq \nu^{-(\nu-b)/b} B_p \|\theta\|_{2,b}^\nu$ must both be equalities. For the first, write $y = A/\|A\|_{1/s}$; by Lemma 2, $h(\theta) \leq \|A\|_{1/s} \min_{\delta\neq 0} \sum_j y_j \kappa_j(\delta) \leq \|A\|_{1/s} \Gamma_s(p)$, so equality forces $\min_\delta \sum_j y_j \kappa_j(\delta) = \Gamma_s(p)$: y is a maximizer of equation 2, which is condition 1 with $c = \|A\|_{1/s}$. For the second, the first inequality of equation 6 is an equality. It is a sum over j of the inequalities $\sum_i |\beta_{ij}|^r \geq \nu^{1-r} A_j^r$, so each of these is an equality. At a frequency with $A_j > 0$, the equality case of Lemma 8 ($r < 1$) is condition 2. At a frequency with $A_j = 0$ it forces $\beta_{ij} = 0$ for every i: no neuron carries an unused frequency. □

B.3 Proof of Proposition 10

Identity. Write $\vartheta_t = 2\pi j a_t/p$ and $\vartheta_c = 2\pi j c/p$, and expand each cosine into exponentials. Neuron l contributes

$$2^{-\nu}\lambda^\nu \sum_{\varepsilon\in\{\pm1\}^n,\ \varepsilon_c=\pm1} \exp\big(i(\varepsilon\cdot\vartheta + \varepsilon_c\vartheta_c)\big)\, \exp\Big(i\alpha_l\big(\textstyle\sum_t \varepsilon_t + n\varepsilon_c\big)\Big).$$

With $N_- = |\{t : \varepsilon_t = -1\}|$ we have $\sum_t \varepsilon_t = n - 2N_-$, so the l-dependent phase is $\exp(2\pi i\, lu/\nu)$ with $u = n - N_-$ if $\varepsilon_c = +1$ and $u = -N_-$ if $\varepsilon_c = -1$. Summing over $l = 0, \ldots, n$ gives $\nu\,[u \equiv 0 \bmod \nu]$. As $0 \leq N_- \leq n$, this happens only for $(N_- = n, \varepsilon_c = +1)$ and $(N_- = 0, \varepsilon_c = -1)$, i.e. for $e^{\mp i(\sum_t \vartheta_t - \vartheta_c)}$, whose sum is $2\cos(\sum_t \vartheta_t - \vartheta_c)$. The block's table is therefore $\nu\lambda^\nu 2^{-n} \cos\big(2\pi j(a_1 + \cdots + a_n - c)/p\big)$. For the cost, $\sum_a \cos^2(2\pi j a/p + \alpha) = p/2$ for $j \not\equiv 0$, so each of the ν weight vectors of neuron l has squared norm $\lambda^2 p/2$.

Attainment. Let $r = b/\nu$ and $s = 1/r$. Scaling y^* up increases the objective, so $\sum_j (y_j^*)^r = 1$. With $\nu\lambda_j^\nu 2^{-n} = y_j^*$ the network's table is exactly $f(\theta, x)[c] = \sum_j y_j^* \cos\big(2\pi j(y(x) - c)/p\big)$. It depends only on $y(x) - c$, so it is interference-free, and on every input $f(\theta, x)[y(x)] - f(\theta, x)[y(x) - \delta] = \sum_j y_j^* \kappa_j(\delta)$; hence $h(\theta) = \min_{\delta\neq 0} \sum_j y_j^* \kappa_j(\delta) = \Gamma_s(p)$. For any non-negative allocation y, the same construction has penalty

$$P_b := \|\theta\|_{2,b}^b = \sum_j \nu(\nu p/2)^{b/2} (2^n y_j/\nu)^r = \nu^{1-r} B_p^{-r} \sum_j y_j^r.$$

For $y = y^*$ the last sum is one. Thus $\|\theta\|_{2,b}^\nu = P_b^{1/r} = \nu^{s-1} B_p^{-1}$, and

$$\frac{h(\theta)}{\|\theta\|_{2,b}^\nu} = \nu^{1-s} B_p \Gamma_s(p),$$

as claimed. More generally, any y with $\sum_j y_j^r = 1$ gives normalized margin $\nu^{1-s} B_p \min_{\delta \neq 0} \sum_j y_j \kappa_j(\delta)$. □

B.4 Proofs of Propositions 11, 12 and 13

Throughout, $s > 1$, $r = 1/s < 1$, and $\phi(v) := \sum_j v_j^r$.

Reduction to $\mathcal{P}_p$. If y is feasible for equation 2 with worst coverage $m > 0$, then $v = y/m \in \mathcal{P}_p$ and $\phi(v) = m^{-r} \sum_j y_j^r \leq m^{-r}$. Conversely, for $v \in \mathcal{P}_p$ the vector $y = v/\phi(v)^s$ is feasible with worst coverage at least $\phi(v)^{-s}$. Hence $\Gamma_s(p) = (\min_{\mathcal{P}_p} \phi)^{-s}$. A maximizer has $\sum_j y_j^r = 1$, since scaling up raises the objective. So y is a maximizer if and only if $y = \Gamma_s(p)\, v$ with v a minimizer of ϕ over $\mathcal{P}_p$.

Proof of Proposition 11.

(i) Minimizers are vertices. The polyhedron $\mathcal{P}_p$ is nonempty and closed. Since $v_j \leq \phi(v)^{1/r}$ for every j, the sublevel sets of ϕ in $\mathcal{P}_p$ are compact, so a minimum exists. If a minimizer v were not a vertex, it would be the midpoint of distinct $v', v'' \in \mathcal{P}_p$. Strict concavity of $t \mapsto t^r$ on $[0, \infty)$ would give $\phi(v) > \frac{1}{2}(\phi(v') + \phi(v'')) \geq \min \phi$, a contradiction.

(ii) Finitely many switches. The vertex set is finite and does not depend on s. For two vertices v, v', put $u = 1/s$. Then $\phi(v) - \phi(v') = \sum_{k=1}^{N} c_k e^{\lambda_k u}$, where the λ_k are the distinct logarithms of the non-zero entries of v and v', and $c_k \in \mathbb{Z}$ is the difference of their multiplicities.

- If every $c_k = 0$, the multisets agree and the two vertices tie for every s.
- Otherwise the function has at most $N - 1$ real zeros. Induction on N: after multiplying by $e^{-\lambda_1 u}$, differentiation removes one term, and by Rolle's theorem the number of zeros drops by at most one.

With finitely many pairs, the set of minimizing vertices (up to such ties) is constant between finitely many values of s.

(iii) Consequences. The support and the ratios of $y = \Gamma_s(p) v$ are those of v, and $y_j^r = \Gamma_s(p)^r v_j^r = v_j^r/\phi(v)$. □

Proof of Proposition 12. Let $v = y/\Gamma_s(p)$, a vertex of $\mathcal{P}_p$ by Proposition 11, with support S of size K. The active constraints at a vertex have rank $|F|$. The constraints $v_j = 0$ for $j \notin S$ involve only coordinates outside S, so the active coverage rows, restricted to S, have rank K. Choose K independent rows. They give $\sum_j y_j \kappa_j(\delta_i) = \Gamma_s(p)$ and determine y_S uniquely. Since the rows at δ and $-\delta$ coincide, the chosen offsets can be taken in $\{1, \ldots, (p-1)/2\}$.

The block network has table $\sum_j y_j \cos(2\pi j(y(x) - c)/p)$, so its gap at offset δ is $\sum_j y_j \kappa_j(\delta)$ on every input. Thus all $2K$ candidates at offsets $\pm\delta_i$ attain the minimum gap. □

Proof of Proposition 13. Removing frequency j decreases every offset's coverage by at most $2y_j$, proving the lower bound. For the upper bound, if $w_j = 1$ then $m_{-j} = 0$. Otherwise, set $y_j = 0$ and multiply the remaining amplitudes by $(1 - w_j)^{-s}$. The resulting allocation has budget one, so its worst coverage obeys $(1 - w_j)^{-s} m_{-j} \leq \Gamma_s(p) \leq m/(1 - \varepsilon)$. Finally, $(1 - w)^s \leq 1 - w$ and Bernoulli's inequality $(1 - w)^s \geq 1 - sw$ give $w \leq 1 - (1 - w)^s \leq sw$. □

B.5 Proof of Theorem 14 and Corollary 15

Write $F_y(\delta) := \sum_{j \in F} y_j \kappa_j(\delta)$ and $m(y) := \min_{\delta \neq 0} F_y(\delta)$. Let $r = 1/s$ and $a_s = 3 + s/2$; logarithms are natural. The constants below are chosen for simplicity, not sharpness.

Lower bound. Let $L = \lceil 8 \ln p \rceil$ and draw $J_1, \ldots, J_L$ independently and uniformly from F. For fixed $\delta \neq 0$,

$$\mathbb{E} \cos(2\pi J_1 \delta/p) = \frac{2}{p-1} \sum_{j \in F} \cos(2\pi j \delta/p) = -\frac{1}{p-1}.$$

By Hoeffding's inequality for variables in $[-1,1]$ (Hoeffding, 1963), $\Pr\left[\frac{1}{L}\sum_\ell \cos(2\pi J_\ell \delta/p) > \frac{1}{2}\right] \le e^{-L/8} \le \frac{1}{p}$. A union bound over the $p-1$ offsets leaves positive probability that $\sum_\ell \kappa_{J_\ell}(\delta) \ge L/2$ for every $\delta \neq 0$; fix such a draw. Let n_j be the multiplicity of j in the draw and set $y_j = n_j L^{-s}$. Then

$$q := \sum_j y_j^r = L^{-1}\sum_j n_j^r \le 1, \qquad F_y(\delta) = L^{-s}\sum_\ell \kappa_{J_\ell}(\delta) \ge \tfrac{1}{2}L^{1-s}.$$

Here $q > 0$. Replacing y by y/q^s makes the budget exactly one, preserves its support, and can only increase coverage. Thus the lower bound is achieved by a feasible allocation using at most L frequencies. Since $L \le 9\ln p$ for $p \ge 3$, also $\Gamma_s(p) \ge \frac{1}{2}(9\ln p)^{1-s}$. This is the cyclic case of the random Cayley graph argument of Alon & Roichman (1994).

A positive-Fourier inequality. If every non-zero offset had large coverage, the average of $e^{-tF_y(\delta)}$ would be small. Positivity of Fourier coefficients forces the same average to remain large under a bounded amplitude budget; comparing the two estimates will bound the margin.

Let $I_0(z) = \frac{1}{\pi}\int_0^\pi e^{z\cos\vartheta}d\vartheta$ and $\psi(z) = z - \ln I_0(z)$. For $z \ge 0$, the identity

$$e^{z\cos\vartheta} = \exp\left((z/2)e^{i\vartheta}\right)\exp\left((z/2)e^{-i\vartheta}\right)$$

and the absolutely convergent exponential series show that all Fourier coefficients are non-negative, with constant coefficient $I_0(z)$. Expand each factor of $e^{-tF_y(\delta)} = \prod_j e^{-ty_j}e^{ty_j\cos(2\pi j\delta/p)}$ and average over $\delta \in \mathbb{Z}_p$. The average keeps the products whose frequencies sum to $0 \bmod p$. All are non-negative, and they include the product of the constant terms. Hence, for $t \ge 0$,

$$\frac{1}{p}\sum_{\delta\in\mathbb{Z}_p} e^{-tF_y(\delta)} \;\ge\; \prod_j e^{-ty_j}I_0(ty_j) = \exp\Big(-\sum_j \psi(ty_j)\Big). \tag{7}$$

No independence of the phases $j\delta$ is used: resonances only add non-negative terms.

A scalar bound. We claim $0 \le \psi(z) \le a_s \ln(1+z^r)$ for $z \ge 0$.

- Non-negativity: $I_0(z) \le e^z$.
- For $z \le 1$: $I_0 \ge 1$ gives $\psi(z) \le z$. Also $\ln(1+z^r) \ge z^r/2 \ge z/2$ and $a_s \ge 2$.
- For $z \ge 1$: integrating only over $\vartheta \le z^{-1/2}$ and using $\cos\vartheta \ge 1 - \vartheta^2/2$ gives $I_0(z) \ge e^{z-1/2}/(\pi\sqrt{z})$, so $\psi(z) \le \ln\pi + \frac{1}{2} + \frac{1}{2}\ln z$. Meanwhile $3\ln(1+z^r) \ge 3\ln 2 > \ln\pi + \frac{1}{2}$, and $\frac{s}{2}\ln(1+z^r) \ge \frac{1}{2}\ln z$.

If $K \ge 1$ and y has at most K non-zero entries and $\sum_j y_j^r \le 1$, concavity of the logarithm then gives

$$\sum_j \psi(ty_j) \;\le\; a_s\sum_j \ln(1+t^r y_j^r) \;\le\; a_s K\ln(1+t^r/K) \;\le\; a_s t^r. \tag{8}$$

Upper bounds. Since $F_y(0) = 0$ and $F_y(\delta) \ge m(y)$ otherwise,

$$\frac{1}{p}\sum_\delta e^{-tF_y(\delta)} \le \frac{1}{p} + e^{-tm(y)}. \tag{9}$$

- Take $t = (\ln p/(2a_s))^s$. By equation 7–equation 9, $e^{-tm} \ge p^{-1/2} - p^{-1} \ge p^{-1/2}/3$, so $m(y) \le (\frac{1}{2}\ln p + \ln 3)/t \le \frac{3}{2}(2a_s)^s(\ln p)^{1-s}$ for every feasible y.
- Keeping K in equation 8 and taking $t^r = K(p^{1/(2a_sK)} - 1)$ instead gives, for every feasible y with at most K non-zero entries,

$$m(y) \;\le\; \tfrac{3}{2}(\ln p)^{1-s}\Big(\frac{x}{e^{x/(2a_s)}-1}\Big)^s, \qquad x = \ln p/K. \tag{10}$$

Number of frequencies.

- Lower bound. A maximizer has $m = \Gamma_s(p) \ge \frac{1}{2}9^{1-s}(\ln p)^{1-s}$. With equation 10 and $e^z - 1 \ge z^2/2$ this gives $\frac{1}{2}9^{1-s} \le \frac{3}{2}(8a_s^2/x)^s$, i.e. $K = \ln p/x \ge c_s \ln p$.

- Upper bound. Proposition 13 with $\varepsilon = 0$ gives $m - 2y_j \le (1-w_j)^s m$. So $2y_j \ge m(1-(1-w_j)^s) \ge m\, w_j = m\, y_j^r$, i.e. $w_j \ge (m/2)^{1/(s-1)}$. As $\sum_j w_j = 1$, $K \le (2/m)^{1/(s-1)} \le 4^{1/(s-1)} L \le 9 \cdot 4^{1/(s-1)} \ln p$, and every share is at least $4^{-1/(s-1)}/L \ge 4^{-1/(s-1)}/(9 \ln p)$. Keeping $L = \lceil 8 \ln p \rceil$ gives the dense-regime bound in the main text whenever $|F| > L$.

Networks. Write $\gamma^*_{b,m}(p)$ for the maximum margin at width m. The universal bound gives $\gamma^*_{b,m}(p) \le B_p \Gamma_s(p)$ for every m. The normalized random allocation above uses at most L frequencies, so the block construction gives, for every $m \ge \nu L$,

$$\tfrac{1}{2} \nu^{1-s} B_p L^{1-s} \le \gamma^*_{b,m}(p) \le B_p \Gamma_s(p).$$

Together with the programme upper bound and $B_p = 2(2p\nu)^{-\nu/2}$, this proves the stated order uniformly over these widths. If m is also large enough to realize a programme maximizer, Theorem 9 gives the stronger lower bound $\nu^{1-s} B_p \Gamma_s(p)$.

Corollary 15. If $A = 0$, Lemma 2 gives $h(\theta) \le 0$, so the claim is immediate. Otherwise $y = A/\|A\|_{1/s}$ has budget one and at most K non-zero entries. Lemma 2, the budget estimate in Theorem 4, and equation 10 give

$$h(\theta) \le \|A\|_{1/s} m(y), \qquad \|A\|_{1/s} \le C_b^{\nu/b} \|\theta\|_{2,b}^{\nu},$$

which prove the displayed bound. For product networks, $C_b^{\nu/b} = B_p$. At any comparison width $m \ge \nu L$, the lower bound just proved shows that the ratio to $\gamma^*_{b,m}(p)$ is at most

$$3(9\nu)^{s-1} \Big(\frac{x}{e^{x/(2a_s)} - 1} \Big)^s.$$

This tends to zero as $x = \ln p / K \to \infty$. □

B.6 Further remarks on Section 4

Equal shares. Restricting equation 2 to equal shares on a set of K frequencies, with $M_K(p) := \max_{|S|=K} \min_{\delta \ne 0} \sum_{j \in S} \kappa_j(\delta)$, gives $\Gamma_s(p) \ge \max_K K^{-s} M_K(p)$, a cheap approximation. Since the worst offset is no better than the average, $M_K \le pK/(p-1)$, so $K^{-s} M_K \le \frac{p}{p-1} K^{1-s}$ decays in K for $s > 1$. In Table 1 the equal-share argmax agrees with K^* except at $(p, \nu) = (59, 3)$, where it gives 6, and $(113, 3)$, where it gives 5; where adjacent equal-share values are within a few percent, the honest statement is a band of width one.

Shares in the block. Optimal shares are not equal (at $p = 59$, $s = 3/2$ they range over 0.15–0.25), and a ν-block with equal amplitudes on the optimal support does not attain Theorem 9.

Weaker cancellation suffices. The proof of Lemma 8 uses only the $\binom{\nu}{2}$ interval patterns $T = \{t' + 1, \dots, t\}$ of equation 3.

Other networks. The upper bound of Theorem 9 uses only the single-neuron budget and the sign rule. For the power network it holds with B_p replaced by $n!\, B_p$; the product-network attainment construction does not establish sharpness for the power network (Appendix D). For a network whose neurons satisfy the sign rule with n operand factors, the factor $\nu^{-(\nu-b)/b}$ becomes $(n+1)^{-(\nu-b)/b}$: it counts factors, not degree.

Scope. (i) Lemma 1 averages over every input; with a training fraction below one, the training-set margin need not obey the bound. (ii) Interference-freeness is a condition on the network, not a fact about trained networks, which carry interference at the sign-flipped modes (Section 6). (iii) K^* is the support size of a maximizer of equation 2.

C The single-neuron budget and the sign rule for concrete networks

Product neuron. $\phi(\{u_1, \dots, u_n, w\}, x) = u_1(a_1) \cdots u_n(a_n)\, w$, $\nu = n + 1$. By Appendix A.1, $\beta_j = 2p^{-\nu} \prod_t \hat{u}_t(j)\, \overline{\hat{w}(j)}$. The coefficient at the mode flipping T is

$2p^{-\nu}\prod_{t\notin T}\hat{u}_t(j)\prod_{t\in T}\hat{u}_t(-j)\,\overline{\hat{w}(j)}$, and $\hat{u}_t(-j)=\xi_{tj}\hat{u}_t(j)$ with $\xi_{tj}=\overline{\hat{u}_t(j)}/\hat{u}_t(j)$ (:= 1 if $\hat{u}_t(j)=0$); this is Lemma 7. For the budget, let $m_j=\sum_t|\hat{u}_t(j)|^2+|\hat{w}(j)|^2$. AM–GM over the ν factors gives $|\prod_t\hat{u}_t(j)\hat{w}(j)|\le(m_j/\nu)^{\nu/2}$. Since $b/2\ge 1$, $t\mapsto t^{b/2}$ is superadditive, so $\sum_j m_j^{b/2}\le(\sum_j m_j)^{b/2}$; and Parseval gives $\sum_{j\in F}m_j\le\frac{p}{2}\|\omega\|_2^2$. Hence

$$\sum_{j\in F}|\beta_j(\omega)|^{b/\nu}\le(2p^{-\nu})^{b/\nu}\Big(\frac{p}{2\nu}\Big)^{b/2}\|\omega\|_2^b,$$

$$C_b^{\nu/b}\le 2p^{-\nu}\Big(\frac{p}{2\nu}\Big)^{\nu/2}=B_p.$$

For a non-zero neuron, equality in the budget inequality holds if and only if it is balanced (all ν Fourier magnitudes equal at every $j\in F$), has no constant component, and, for $b>2$, uses a single frequency; at $b=2$ the superadditivity step is an identity, so a neuron may carry several frequencies at no extra cost. The zero neuron also gives equality. A neuron whose ν weight vectors are cosines of one frequency with equal norms attains the budget, so $C_b^{\nu/b}=B_p$.

Power neuron. $\phi(\{u_1,\dots,u_n,w\},x)=\big(\sum_t u_t(a_t)\big)^n w$, $\nu=n+1$. Expand the power into monomials $\prod_t u_t(a_t)^{e_t}$ with $\sum_t e_t=n$. A monomial with some $e_t=0$ has no Fourier content in slot t, and one with some $e_t\ge 2$ has some other $e_{t'}=0$; so only the multilinear monomial, with coefficient $n!$, reaches a mode with all ν entries non-zero. Hence $\beta_j=n!\cdot 2p^{-\nu}\prod_t\hat{u}_t(j)\,\overline{\hat{w}(j)}$, Lemma 7 holds with the same ξ_{tj}, and $C_b^{\nu/b}=n!\,B_p$: the upper bound follows from the product budget, and a balanced single-frequency cosine neuron attains it. In particular $C_3=2B_p$ at $n=2$. This is a single-neuron budget identity, not a network-level attainment claim. Theorem 4 and the upper bound of Theorem 9 apply. Attainment requires additional control of the monomials with a zero operand slot: Definition 6 permits them, but they can reduce the worst-case margin. Consequently the product-block identity alone does not give an attaining power network.

Where the approach does not apply. A network with a shared linear layer in front of the neurons is not a sum of neurons each paying its own norm: the shared layer is priced by a nuclear norm that rewards carrying many frequencies in one neuron, and the single-neuron budget does not bound it. Such networks are outside this paper.

D THE GLOBAL CONJECTURE, AND WHAT IS KNOWN

Conjecture 19. *For odd prime p, $n\ge 1$ and $2\le b\le\nu$, the maximum $L_{2,b}$-margin over* all *product networks equals the interference-free value of Theorem 9: no network carrying interference exceeds it.*

Proved here. At $b=\nu$, Corollary 5 and Proposition 10 give matching unrestricted upper and lower bounds (at sufficient width). For $2\le b<\nu$, Theorem 14 determines the order of the unrestricted optimum, while Theorem 9 gives the exact interference-free value. Equality of these two optima throughout this range remains open.

Power networks. The universal and interference-free upper bounds apply with single-neuron constant $C_b^{\nu/b}=n!B_p$. Their attainment is a separate question from the product-network result; the quadratic endpoint $n=2$, $b=3$ is covered by Corollary 5.

b *outside* $[2,\nu]$. For $b>\nu$ the normalized margin is unbounded as the width grows. For $b<2$ the superadditivity step of Appendix C fails for the product network (one neuron carrying every frequency exceeds it by $|F|^{1-b/2}$). The available bound is the $b=2$ problem with a rescaled budget, which is not attained, so nothing about the number of features is claimed there.